

The University of Pennsylvania Glioblastoma (UPenn-GBM) Cohort

Advanced MRI, clinical, genomics, and radiomics - Bakas et al. - Scientific Data - 2022

Purpose. UPenn-GBM is a comprehensive public data resource designed to support reproducible computational and translational glioblastoma research by pairing multimodal MRI with clinical, molecular, segmentation, diffusion/perfusion, and radiomic information.

1. Scope of the dataset

Category	What is included
Patients	630 patients diagnosed with de novo GBM; records identified at UPenn from 2006-2018
Time points	611 baseline pre-operative scans; 60 follow-up scans before second surgery, with 19 follow-up-only cases
Structural MRI	T1, T1-Gd, T2, and T2-FLAIR
Advanced MRI	DTI for 592 scans (88.2%); DSC perfusion for 534 scans (79.6%)
Clinical / molecular	Age, sex, resection status, survival, KPS where available, IDH1 and MGMT status
Derived data	Standardized MRI, diffusion/perfusion maps, expert-refined tumor labels, and radiomic features

2. A multimodal research resource

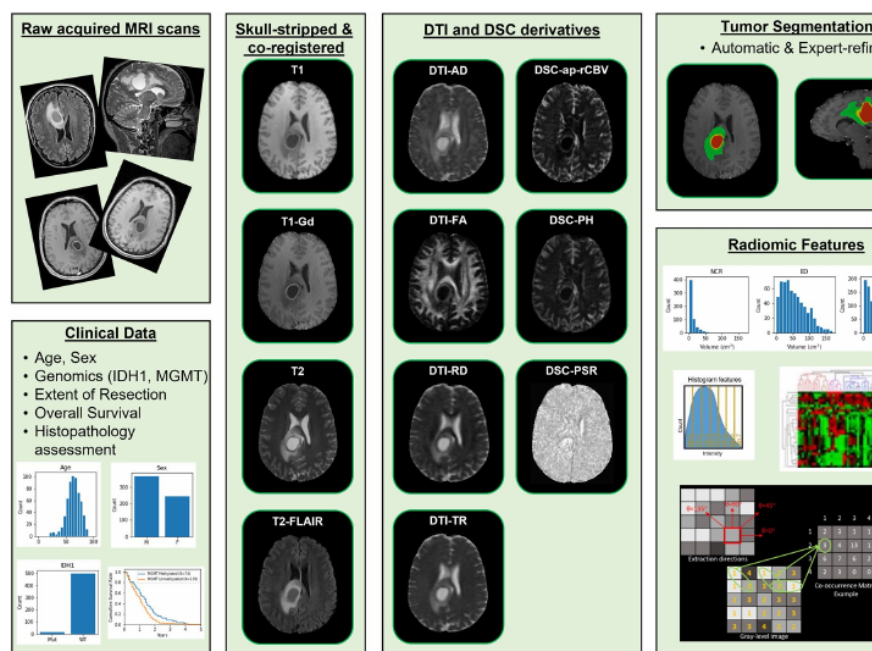


Fig. 1 Visual summary of the “University of Pennsylvania Glioblastoma Advanced Imaging, Clinical.

Article Figure 1 (p. 2). Raw MRI, harmonized structural scans, DTI and DSC derivatives, clinical/genomic data, expert-refined tumor segmentation, and radiomic features.

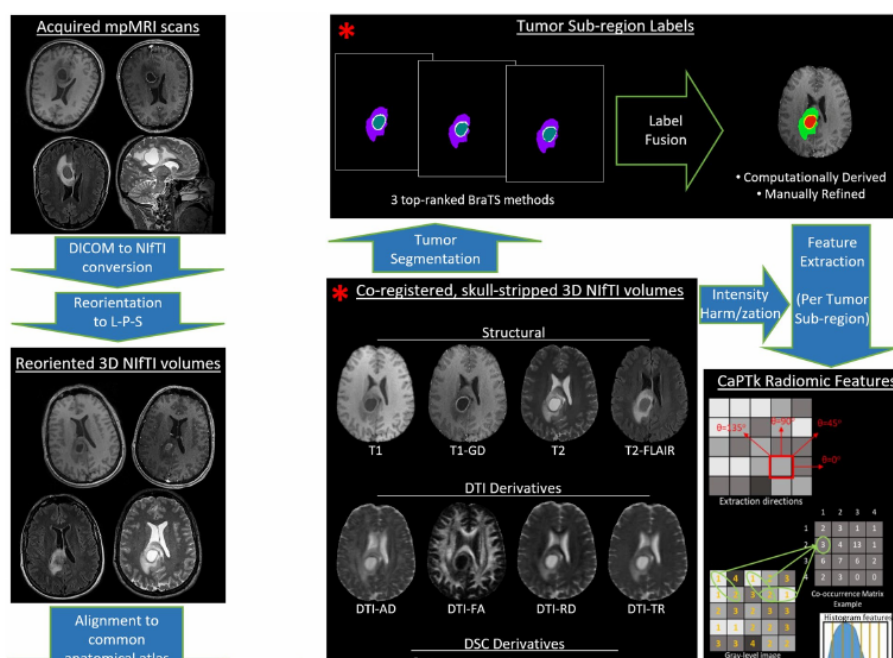
The resource addresses common limitations of public GBM datasets - cohort size, inconsistent acquisition, variable quality, and sparse clinical or molecular metadata. It supports predictive, prognostic, diagnostic, radiogenomic, segmentation and radiomics studies.

3. Clinical and molecular composition

The cohort is 60% male and 40% female; 58.3% are age 50-69 and 28.6% are 70+. Among baseline cases, 362 underwent gross-total resection and 211 partial resection. Overall survival is available for 452 patients. IDH was wildtype in 499, mutated in 16, and NOS/NEC in 96. MGMT status is available for 317 baseline cases: 140 methylated and 177 unmethylated.

4. Harmonized image preprocessing

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Article Figure 2 (p. 5). Harmonized preprocessing: DICOM-to-NIfTI conversion, reorientation, rigid registration through T1-Gd to a common atlas, 1-mm isotropic resampling, defacing, skull stripping and downstream preparation.

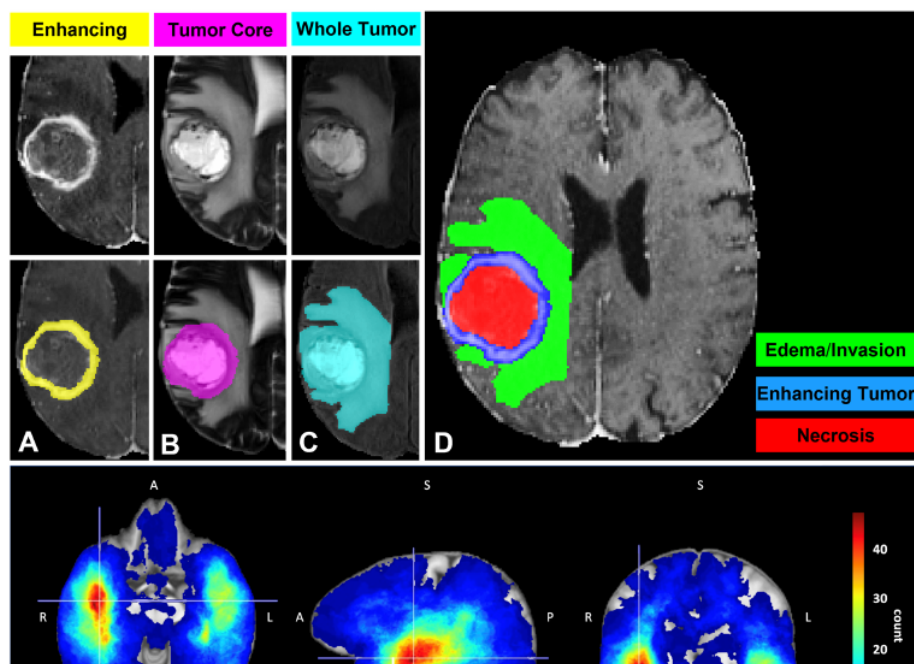
The protocol follows BraTS conventions and supplies both original and standardized data. Registration was manually reviewed, defacing masks were quality-checked (about 10% required refinement), and brain extraction used the glioma-aware deep-learning BrainMaGe approach.

5. Tumor segmentation

Enhancing tumor (ET), necrotic/cystic tumor core (NCR), and peritumoral edematous/infiltrated tissue (ED) were generated by label fusion of high-performing deep-learning methods including DeepMedic, DeepSCAN and nnU-Net. A subset of 232 subjects was expert-evaluated and manually refined when needed.

6. Imaging labels and advanced MRI derivatives

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Article Figure 3 (p. 6). Examples of enhancing tumor, tumor core, whole tumor and separate ET/NCR/ED labels superimposed on MRI, plus cohort-wide tumor-core spatial distribution.

DTI supports diffusion-derived measures associated with tissue microstructure and density, while DSC MRI supports perfusion derivatives. Diffusion maps were checked for brain coverage, registration and physically valid values; DSC time curves were inspected for adequate bolus sampling before derivative calculation.

7. Radiomics

CaPTk-derived radiomic features span intensity statistics, histogram descriptors, volumetric/morphologic measures, and texture families including GLCM, GLRLM, GLSZM, NGTDM and local binary patterns. They are provided 'as-is' to facilitate downstream studies rather than as prevalidated biological biomarkers.

8. Technical validation and reproducibility

Manual quality control is a major feature of the release. Registrations were inspected and corrected; defacing masks reviewed; brain masks and tumor labels checked and refined; and perfusion/diffusion derivatives subjected to modality-specific validity checks. The authors nevertheless acknowledge uncertainty in visually defining exact tissue and tumor boundaries.

9. What the dataset enables

Research area	Examples
Segmentation	Training and benchmarking ET/NCR/ED and whole-tumor models
Radiomics / radiogenomics	Linking quantitative phenotype with IDH, MGMT and other endpoints
Outcome modeling	Overall-survival and prognostic stratification
Recurrence / progression	Recurrence-location prediction and progression vs pseudoprogression
Advanced MRI	Diffusion- and perfusion-based tissue characterization
Reproducibility	Comparative algorithm development using standardized public data

Subsets of the cohort had already supported studies of recurrence, molecular subtype, survival, BraTS/FeTS segmentation, pseudoprogression, imaging subtypes and brain extraction, illustrating the breadth of downstream applications.

10. Bottom line

UPenn-GBM is more than an MRI image set. It is an integrated **imaging + clinical + molecular + segmentation + radiomics** resource with standardized preprocessing and substantial quality control. For brain-tumor AI, it bridges pure image segmentation with clinically meaningful prediction, radiogenomics, advanced MRI analysis and reproducible multimodal modeling.